

1. Course number and name

CHE 341. Chemical Process Control

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Michael Caracotsios

4. Textbook, title, author, publisher, year

Process Dynamics and Control, D.E. Seborg, T.F. Edgar, D.A. Mellichamp, and F.J. Doyle, Wiley, 2011.

a. other supplemental materials

- Chemical Process Control Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020
- T.E. Marlin, Process Control: Designing Processes and Control Systems for Dynamic Performance, McGraw-Hill, 2000; B.A. Ogunnaike and W.H. Ray, Process Dynamics, Modeling and Control, Oxford, 1994; J.B. Riggs and M. N. Karim, Chemical and Bio-Process Control, Ferret Publishing, Austin, Texas, 2016; G. Stephanopoulos, Chemical Process Control: An Introduction to Theory and Practice, Prentice Hall, 1984.
- SOFTWARE: Athena Visual Studio, EXCEL based Thermodynamic Calculator CHE301 Calculator.xlsm; EXCEL based Process Control tool CHE341 Loop Simulator.xlsm

5. Specific course information

a. Brief description of the content of the course (catalog description)

Analysis and design of chemical processes and control systems. Feedback and feedforward controllers. Stability, tuning, and simulation of P-I-D controllers. Introduction to the control of chemical plants and digital process control.

b. Prerequisites or co-requisites

MATH 220 and CHE 312 and CHE 313 and CHE 321.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand the basic concepts of process dynamic modeling for a variety of chemical engineering systems.
- Know how to develop empirical models from process data using the step response methodology
- Understand the theory of feedback and feed forward control and determine optimal controller settings for stable and robust performance.
- Know when to select and how to implement cascade and feed forward controllers
- Apply available computer software (Matlab, Excel or Athena Visual Studio), or use Laplace transformation theory to solve process dynamic models and

investigate their open and closed loop response to various external disturbances.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.			X	

7. Brief list of topics to be covered

INTRODUCTION TO PROCESS CONTROL IN CHEMICAL ENGINEERING: Elements of Differential and Integral Calculus, Numerical Methods for Algebraic and Differential Equations, Mass and Energy Balances in the Context of Process Control, Tutorials on Programming with VBA and Athena Visual Studio

CONTROL OF INDIVIDUAL UNITS: Reactors, Heat Exchangers, Distillation Columns, Compressors, Other Unit Operations

CONTROL CONCEPTS AND DYNAMIC MODELING: Development of Dynamic Process Models, Solution of Dynamic Models and Open Loop Response, Laplace Transforms and Transfer Functions, Linearization of Process Dynamic Models, Process Identification from Plant Data

FEEDBACK CONTROL FUNDAMENTALS: P, PI and PID Controllers, Block Diagrams and Routh-Hurwitz Stability Criterion, Instrumentation Hardware and Representation

FEEDBACK CONTROLLER DESIGN AND TUNING: Criteria for Controller Tuning, PID Controller Tuning Methods, Guidelines and Troubleshooting, Common Control Loops

ADVANCED CONTROL SYSTEMS AND MIMO SYSTEMS: Feedforward and Cascade Control, Interaction and Loop Pairing and Decoupling, Relative Gain Array Control of Common Industrial Process Units, Model Predictive Control of Linear Processes, Advanced Process Control and Real-Time Optimization